Module Overview

Essential Question: What does it take to make a successful invention?
Quote: "To invent, you need a good imagination and a pile of junk." — Thomas Edison
Big Idea Words (4): invention, brilliant, productive, original
Main Selections (workbook 4):

- Timeless Thomas: How Thomas Edison Changed Our Lives · Gene Barretta · Biography / Informational Text
- A Bumpy Ride · Sharon Katz Cooper & Rachel Young (illus. Ariane Elsammak) · Informational Text
- Rosie Revere, Engineer · Andrea Beaty (illus. David Roberts) · Narrative Poem
- Edison's Best Invention · (HMH, no byline) · Opinion Text

Note. The SoT lists 26 core vocabulary items on Worksheet A (4 Big Idea + 6 Timeless Thomas + 4 A Bumpy Ride + 6 Rosie Revere + 6 Edison's Best Invention), in myBook appearance order. This G3 Teacher's Guide labels each selection's panel "Critical Vocabulary" — the string "Power Words" does not appear in the source. The teacher "Words for Using ..." strategy boxes are NOT counted as core vocabulary. "A Century of Amazing Inventions" is the Big-Idea Short Read (no Critical Vocabulary panel) and carries the 4 Big Idea Words; it is excluded from the per-selection workbook units. Worksheet E is intentionally omitted — this workbook series uses 4 worksheets (A·B·C·D). Module 8 has NO video/media selection among its numbered selections.

Answer Distribution (78 Multiple-Choice Items)

Verified counts across Worksheets A (26), B (32), and D (20). Worksheet C is open-response IB and not counted in MCQ balance.

A · 20 · B · 20 · C · 19 · D · 19

These four totals are the count of correct-answer LETTERS (A/B/C/D) across all 78 multiple-choice items — 20 answers are "A", 20 are "B", 19 are "C", 19 are "D" (as even as 78 allows). Balance enforced via choice-order rotation only; correct-answer meanings and option texts are preserved from the Source of Truth. All 26 SoT vocabulary words covered on Worksheet A in myBook appearance order (Big Idea → Timeless Thomas → A Bumpy Ride → Rosie Revere → Edison's Best Invention).

Voca Trap Items (Worksheet A)

8 of 26 ≈ 31%, all flagged below. Trap pattern: polysemy / part-of-speech / sound-alike / near-distractor baits drawn from the natural meaning field of each word — no hallucination.

- Q3 **productive** — opposite-meaning bait ("waste most of your time") vs. "able to do a lot with the time you have"
- Q7 **breakthrough** — literal "break-through" bait (break a window / run through a wall) vs. "make an important discovery"
- Q13 **craze** — sound-alike bait ("cracks" / "crazy, dangerous") vs. "very popular for a period of time"
- Q16 **perplexed** — opposite-meaning bait ("calm and perfectly sure") vs. "confused and worried"
- Q18 **lingers** — opposite-meaning bait ("disappears right away") vs. "remains for a long time"
- Q19 **whirled** — sound/sense bait ("whirring noise but stayed still") vs. "turned around several times"
- Q22 **compact** — homophone bait ("packed in a box" / "compact disc") vs. "can fit in a small space"
- Q26 **innovative** — "brand new in its packaging" bait vs. "has never been thought of before"

Reading Trap Items (Worksheet B)

4 of 32 ≈ 13%, all flagged below. Trap pattern for Section B: MAIN-idea bait (a true detail mistaken for the main idea) and length-balanced correct answers so a longest-option-wins guess fails. Distractors built from SoT only — no hallucination. Codes used are RI.3.1 / .2 / .3 / .5 / .6 / .8 for the three informational/opinion selections and RL.3.1 / .2 / .3 / .5 / .6 for the one narrative poem. **RI.3.4, RI.3.7, and RL.3.7 are NOT used**, and there are no "from the text" snippet boxes.

- Q8 **Timeless Thomas MAIN idea** — "1,093 patents" / "Black Maria" / "began as a boy" are true details, baited against the real main idea: a curious inventor who never gave up changed the world
- Q16 **A Bumpy Ride MAIN idea** — "Queen Victoria" / "leather caps" / "ordinaries" bait vs. many inventors each improved on the last to make the modern bicycle

- Q24 **Rosie Revere theme** — "cheese hat" / "Aunt Rose built airplanes" / "Lila Greer" bait vs. failure is part of inventing, and the only true failure is quitting
- Q32 **Edison's Best Invention MAIN idea** — "Black Maria" / "Greek roots" / "kinetograph is a camera" bait vs. a good invention inspires new inventions and changes how we live

How the Four Worksheets Build Toward the EQ

This module asks **"What does it take to make a successful invention?"** Each worksheet adds one layer of the answer, so the teacher can move from word knowledge to deep thinking in order:

- **A · Vocabulary** — students first own the four Big Idea Words of invention (invention, brilliant, productive, original) and the words each selection uses (valuable, device, breakthrough, dictation, technology, radar, contraption, quest, craze, era, engineer, perplexed, dynamo, lingers, whirled, baffled, visionary, compact, singles, memorable, system, innovative). Without these words, students cannot describe how an invention is made.
- **B · Reading Comprehension** — students gather evidence from all four selections: a biography of a real inventor (Edison), an informational history (the bicycle), a narrative poem (Rosie Revere), and an opinion text (the light bulb). Structure and point-of-view items prepare the thinking that Section C needs. Three selections are RI; Rosie Revere is RL.
- **C · IB Critical Thinking** — students synthesize: they trace Edison's PERSEVERANCE, the bicycle's step-by-step CHANGE, Rosie's lesson about failure, how a real inventor and a story inventor compare, and finally answer the EQ with two selections. The C activities grow straight out of the B evidence.
- **D · MAP Practice** — four NEW passages let students apply the same idea to inventions they have not studied here (the sticky note, the eraser pencil, the bicycle bell, and Braille), so the EQ becomes a transferable lens about what it takes to invent, not a single-text answer.

Teaching tip. If time is short, the EQ can be reached with Section A (words) + Section C Q5 (synthesis) alone — but Section B supplies the evidence that makes Q5 specific rather than generic.

Source-of-Truth Verification

Into Reading · Grade 3 · Module 8 · For Teacher Use

VERIFIED

How Every Item Is Grounded

Every item on this Answer Key traces to the Source-of-Truth JSON for G3 Module 8, built from the Teaching Pal OCR (4,755 lines, 43 pages) and verified against the PDF intro spreads (every Critical Vocabulary panel was read from a high-resolution crop). No characters, words, or events outside the Source of Truth were added.

- **Vocabulary (Worksheet A):** all 26 words come from the module's Big Idea Words (4: invention, brilliant, productive, original) and the four "Critical Vocabulary" panels — Timeless Thomas 6, A Bumpy Ride 4, Rosie Revere 6, Edison's Best Invention 6. Each meaning is checked against the myBook inline definition. The teacher "Words for Using ..." strategy boxes are deliberately NOT counted as core vocabulary.
- **Terminology:** this G3 Teacher's Guide labels each panel "Critical Vocabulary," not "Power Words" — the workbook uses that exact label, matching the source.
- **Reading (Worksheet B):** the 32 items use RI.3 codes for the three informational/opinion selections (RI.3.1 / .2 / .3 / .5 / .6 / .8) and RL.3 codes for the one narrative poem (RL.3.1 / .2 / .3 / .5 / .6). Each correct answer is grounded in the Source of Truth; distractors are SoT-anchored, never invented. RI.3.4, RI.3.7, and RL.3.7 are not used, and there are no "from the text" snippet boxes.
- **Genre & credits:** Timeless Thomas = Biography / Informational (Gene Barretta); A Bumpy Ride = Informational Text (Sharon Katz Cooper & Rachel Young, illus. Ariane Elsammak); Rosie Revere, Engineer = Narrative Poem (Andrea Beaty, illus. David Roberts); Edison's Best Invention = Opinion Text (HMH, no byline). One biography + one informational + one narrative poem + one opinion — never labeled the same.
- **IB (Worksheet C):** the PYP concepts — Perseverance, Change, Innovation — follow the Source-of-Truth invention/science focus of this module. The Essential Question text on Q5 matches the workbook word for word.
- **MAP (Worksheet D):** four NEW passages (152 / 159 / 160 / 161 words), RIT 195→215, Lexile 410L–680L. None recall the four anchor selections; each applies the EQ to a fresh invention.
- **Answer balance:** 78 multiple-choice items distribute as 20 "A" / 20 "B" / 19 "C" / 19 "D" (as even as 78 allows), achieved by choice-order rotation only — every meaning is preserved from the Source of Truth.

MAP Mini-Passage Word Counts (Worksheet D)

- Passage 1 · The Sticky Note · **152 words** · Lexile 410–460L · RIT 195–201
- Passage 2 · The Better Pencil · **159 words** · Lexile 470–530L · RIT 202–207
- Passage 3 · How the Bicycle Bell Rang · **160 words** · Lexile 540–600L · RIT 208–212
- Passage 4 · The Boy Who Made Braille · **161 words** · Lexile 610–680L · RIT 213–215

Word counts match the workbook passage meta exactly. All four within target band 120–200 words; RIT 195→215; Lexile 410L–680L. Passages are NEW — no recall of the four anchor selections.

What This Module Does NOT Contain (Guardrails)

- No "Power Words" label — the G3 source says "Critical Vocabulary."
- No Worksheet E — this workbook series uses four worksheets (A · B · C · D).
- The module quote IS by a NAMED person, Thomas Edison. It is NOT anonymous.
- "A Century of Amazing Inventions" is the Big-Idea Short Read (a 20th-century inventions timeline, no Critical Vocabulary panel) and is excluded from the per-selection units; it carries the 4 Big Idea Words instead.
- Module 8 has NO video/media selection among its numbered selections. The only video is the module-opener "Get Curious" intro video "Getting There" — it is not a body selection.

- The third selection is "A Bumpy Ride" (about the history of the bicycle) — NOT "A Buggy Ride." Verified on the title spread and by four verbatim body occurrences.
- A Bumpy Ride has only 4 Critical Vocabulary words (contraption, quest, craze, era) — the smallest panel; it is not padded to 6. Counts are 6/4/6/6 = 22 Critical Vocabulary; Big Idea (4) + Critical (22) = 26 total.
- Gene Barretta is the author of Timeless Thomas (selection 1). He is QUOTED inside Edison's Best Invention (selection 4) as evidence, but he is NOT the author of that opinion text — it has no printed byline.
- Performance Task = "Write an Expository Essay." It is not an opinion or narrative essay.

Worksheet A — Vocabulary · Answer Key

26 items · CCSS L.3.4 / L.3.5 / L.3.6

BIG IDEA WORDS

Q1 L.3.6

invention

C

Meaning: an invention is something a person creates that did not exist before. **Source:** SoT Big Idea Word 1 — Vocabulary Card 8.1; highlighted in A Century of Amazing Inventions. *Distractors A/B/D give a copy / a fictional story / a rule.*

Q2 L.3.6

brilliant

A

Meaning: when an idea or person is brilliant, it is extremely clever or skillful. **Source:** SoT Big Idea Word 2. *Distractors B/C/D give dull / heavy / old.*

Q3 L.3.5 TRAP

productive

D

Meaning: if you are productive, you can do a lot with the time and resources you have. **Source:** SoT Big Idea Word 3. **Trap:** A reverses it ("waste your time").

Q4 L.3.6

original

B

Meaning: something original is the first of its kind. **Source:** SoT Big Idea Word 4. *Distractors give borrowed / broken / the last one.*

CRITICAL VOCABULARY · TIMELESS THOMAS

Q5 L.3.4

valuable

A

Evidence: SoT — "If something is valuable, it is useful, helpful, or important" (OCR line 639); Edison's "valuable failures." *Distractors give throw away / metal / too small.*

Q6 L.3.4

device

C

Evidence: SoT — "A device is a tool or machine that has a certain purpose" (OCR line 767); the phonograph was the first device to record sound. *Distractors give a song / a person / a place.*

Q7 L.3.5 TRAP

breakthrough

D

Evidence: SoT — "If you make a breakthrough, you make an important discovery after many tries" (OCR lines 769–770). **Trap:** A/B bait the literal "break through" (window / wall).

Q8 L.3.4

dictation

B

Evidence: SoT — "Dictation describes the act of writing down or recording words that have been spoken" (OCR lines 860–861); phonographs used for dictation in offices. *Distractors give drawing / fixing / selling.*

Q9 L.3.4

technology

A

Evidence: SoT — "Technology is the use of science to invent useful things or to solve problems" (OCR lines 1066–1067); the fluoroscope used X-ray technology. *Distractors give a story / music / farming.*

Q10 L.3.4

radar

C

Evidence: SoT — "Radar is a way to find unseen objects by using radio signals" (OCR lines 1378–1379); Edison discovered radio waves. *Distractors give a light / a music machine / a cutting tool.*

CRITICAL VOCABULARY · A BUMPY RIDE

Q11 L.3.4

contraption

D

Evidence: SoT — "A contraption is a mechanical gadget or device that has a certain purpose" (OCR lines 1993–1994); von Drais's draisienne. *Distractors give a road / a horse / a school.*

Q12 L.3.4

quest

A

Evidence: SoT — "If you go on a quest, you go in search of something important to you" (OCR line 2264); a quest to make two-wheelers safer. *Distractors give wait / a test question / rest.*

Q13 L.3.5 TRAP

craze

C

Evidence: SoT — "If there is a craze for something, it is very popular for a period of time" (OCR lines 2403–2404); cycling became a national craze. **Trap:** A/B bait sound-alikes "cracks"/"crazy."

Q14 L.3.4

era

B

Evidence: SoT — "An era is a period of time in history" (OCR line 2406); bikes ended the era of long dresses. *Distractors give a mistake / a race / a tool.*

CRITICAL VOCABULARY · ROSIE REVERE, ENGINEER

Q15 L.3.4

engineer

D

Evidence: SoT — "An engineer is a person who designs, builds, or operates machines" (OCR lines 2806–2807); Rosie dreamed of becoming a great engineer. *Distractors give painter / teacher / seller.*

Q16 L.3.5 TRAP

perplexed

A

Evidence: SoT — "If you are perplexed, you feel confused and worried about something" (OCR lines 2869–2870). **Trap:** B reverses it ("calm and perfectly sure"); D gives "proud."

Q17 L.3.4

dynamo

C

Evidence: SoT — "A person who is a dynamo has a lot of energy and is very good at what that person does" (OCR lines 2987–2989); Aunt Rose was a true dynamo. *Distractors give tired / never tries / afraid.*

Q18 L.3.5 TRAP

lingers

B

Evidence: SoT — "When something lingers, it remains for a long time" (OCR lines 2990–2991); "time never lingers." **Trap:** A reverses it ("disappears right away").

Q19 L.3.5 TRAP

whirled

D

Evidence: SoT — "Something that whirled turned around several times" (OCR lines 3109–3111); the copter "whirled round and round." **Trap:** A baits the "whirring noise but stayed still" sense.

Q20 L.3.4

baffled

A

Evidence: SoT — "If you are baffled, you are confused" (OCR line 3229); "young Rosie was baffled." *Distractors give cheerful / sleepy / hungry.*

CRITICAL VOCABULARY · EDISON'S BEST INVENTION

Q21 L.3.4

visionary

C

Evidence: SoT — "A visionary is a person who has new or unusual ideas about life in the future" (OCR lines 3715–3716). *Distractors give see in the dark / copies others / wears glasses.*

Q22 L.3.5 TRAP

compact

A

Evidence: SoT — "Things that are compact can fit in a small space" (OCR lines 3962–3963). **Trap:** B baits "packed in a box," D the "compact disc" sound; C reverses it ("large and heavy").

Q23 L.3.4

singles

B

Evidence: SoT — "Music singles are recordings of one song" (OCR lines 3964–3965); without recorded sound there would be no singles. *Distractors give solo singers / book words / big bands.*

Q24 L.3.4

memorable

D

Evidence: SoT — "When something is memorable, it is special enough that people want to remember it" (OCR lines 3966–3967); memorable speeches. *Distractors give easy to forget / too quiet / one language.*

Q25 L.3.4

system

A

Evidence: SoT — "A system is a set of things or ideas that work together to get something done" (OCR lines 3998–3999); a light and power system. *Distractors give one bulb / a story / a digging tool.*

Q26 L.3.5 TRAP

innovative

B

Evidence: SoT — "An innovative idea is one that has never been thought of before" (OCR lines 4001–4002); Edison's innovative grid. **Trap:** A baits "brand new in its packaging"; C gives "copied."

Worksheet B — Reading Comprehension · Answer Key

32 items · CCSS RI.3.1 / .2 / .3 / .5 / .6 / .8 · RL.3.1 / .2 / .3 / .5 / .6

Selection 1 · Timeless Thomas

BIOGRAPHY / INFORMATIONAL · 8 ITEMS

Q1 RI.3.2

B

Evidence: the biography alternates "Present Day" and "Edison's Lab" to show how his inventions still shape modern life (OCR lines 1546–1554). *Others pick a stray detail.*

Q2 RI.3.3

C

Evidence: "he gathered a large team... they started the first research and development laboratory... It became known as the Invention Factory" (OCR lines 620–625). *Others off-text.*

Q3 RI.3.3

A

Evidence: "Edison used his failures as a necessary part of inventing... And he would always try again" (OCR lines 634–637). *Others off-text.*

Q4 RI.3.5

D

Text Structure: SoT — the author uses "Present Day" vs "Edison's Lab" compare-and-contrast spreads (OCR lines 1546–1554). *Others off-text.*

Q5 RI.3.3

A

Evidence: "Edison's tinfoil phonograph was the first device to record sound and play it back. It was a major scientific breakthrough" (OCR lines 743–746). *Others off-text.*

Q6 RI.3.3

B

Evidence: the battery "was originally created to power an electric car. But since the electric car didn't catch on, he found many other... uses for it" (OCR lines 966–970). *Others off-text.*

Q7 RI.3.8

C

Support: "the movie industry was essentially started at Edison's lab... So Edison created the first motion picture camera, the Kinetograph... and built the first movie studio" (OCR lines 1155–1167). *Others true but off-point.*

Q8 RI.3.2 TRAP

D

Main Idea: a curious, persistent inventor who never gave up changed the world with inventions we still use (SoT theme; OCR lines 558–562, 1540–1543). **Trap:** A/B/C are true details, not the main idea.

Selection 2 · A Bumpy Ride

INFORMATIONAL TEXT · 8 ITEMS

Q9 RI.3.2

A

Evidence: the text traces the history of the bicycle through a timeline (1818–1975), each inventor improving on the last (OCR lines 1985–1988, 2512–2517). *Others off-topic.*

Q10 RI.3.3

B

Evidence: von Drais's "contraption, called a draisienne... didn't have pedals. It was more like a sit-on scooter" (OCR lines 1985–1988). *Others off-text.*

Q11 RI.3.3

D

Evidence: "the pedal-powered bike was hard to steer. Macmillan caused the first bike wreck when he plowed into a crowd" (OCR lines 2108–2111). *Others off-text.*

Q12 RI.3.3

C

Evidence: "John Boyd Dunlop wrapped a garden hose around the wheels of his tricycle... air-filled hoses were a success. The modern tire was born" (OCR lines 2367–2374). *Others off-text.*

Q13 RI.3.5

B

Text/Graphic Features: SoT — the timeline "shows the important dates in the history of the bicycle" with a labeled year for each stage (OCR lines 2022–2025, 2204–2215). *Others off-text.*

Q14 RI.3.1

A

Evidence: "to go faster, you needed a bigger wheel (more distance for each pedal). Front wheels got bigger and bigger" (OCR lines 2234–2240). *Others off-text.*

Q15 RI.3.6

C

Author's Craft: SoT — the title fits because "the first bicycles gave people a bumpy ride" AND the history had "many problems, like bumps in a road" (OCR lines 2064–2069). *Others misread it.*

Q16 RI.3.2 TRAP

D

Main Idea: many inventors each improved on the last, and together their ideas made the successful modern bicycle (SoT theme; OCR lines 2512–2517). **Trap:** A/B/C are stray details.

Selection 3 · Rosie Revere, Engineer

NARRATIVE POEM · 8 ITEMS

Q17 RL.3.2

C

Evidence: the poem tells of shy Rosie, who hides her dreams after being laughed at, then learns from her aunt that a "flop" is a "perfect first try" (OCR lines 2797–2800, 3219–3247). *Others off-topic.*

Q18 RL.3.3

D

Evidence: after Uncle Fred "chuckled out loud," Rosie "stuck the cheese hat on the back of her shelf and after that day kept her dreams to herself" (OCR lines 2873–2885). *Others off-text.*

Q19 RL.3.3

A

Evidence: Great-Great-Aunt Rose's wish "to fly" gives Rosie "a daring idea," and "young Rosie knew just how to make her aunt fly" (OCR lines 2983, 3000–3009). *Others off-text.*

Q20 RL.3.3

B

Evidence: "It floated a moment and whirled round and round, then froze for a heartbeat and crashed to the ground" (OCR lines 3098–3101). *Others off-text.*

Q21 RL.3.6

C

Point of View: Aunt Rose says "it crashed. That is true. But first it did just what it needed to do. Before it crashed... it flew!" (OCR lines 3231–3238). *Others reverse her meaning.*

Q22 RL.3.5

D

Story Structure: SoT — the selection is a narrative poem "written in rhymed verse" in four-line aabb stanzas (OCR lines 2806–2816). *Others off-text.*

Q23 RL.3.3

A

Evidence: at Blue River Creek the kids "build gizmos and gadgets... With each perfect failure, they all stand and cheer, but none quite as proudly as Rosie Revere" (OCR lines 3342–3345). *Others off-text.*

Q24 RL.3.2 TRAP

B

Theme: "Life might have its failures, but this was not it. The only true failure can come if you quit" (OCR lines 3246–3247). **Trap:** A/C/D are stray details.

Selection 4 · Edison's Best Invention

OPINION TEXT · 8 ITEMS

Q25 RI.3.2

A

Evidence: "in my opinion, Edison's best invention is the light bulb" (OCR lines 3698–3699). *Others off-topic.*

Q26 RI.3.8

B

Support: "We can compare inventions in two ways. First, good inventions give people ideas that lead to new inventions... Second, good inventions change the way we live" (OCR lines 3701–3708). *Others off-text.*

Q27 RI.3.8

C

Support: "once people had electricity... they could use it to power other inventions... refrigerators, washing machines, and computers" (OCR lines 3989–3993). *Others true but off-point.*

Q28 RI.3.6

D

Author's Craft: the writer cites the book Timeless Thomas as evidence — "According to Gene Barretta... 'the world's first motion picture camera'" (OCR lines 3806–3807, 3934). *Others misread the purpose.*

Q29 RI.3.1

A

Evidence: "Before Edison, people used candles, oil lamps, or gas lights... Burning candles and oil lamps caused many deadly fires" (OCR lines 4106–4109). *Others off-text.*

Q30 RI.3.3

B

Evidence: "Inventors who were inspired by the phonograph gave us vinyl records, compact discs, and digital music" (OCR lines 3939–3940). *Others belong to the light bulb or kinetograph.*

Q31 RI.3.5

C

Text Structure: SoT — the author "compares and contrasts the light bulb to the kinetograph and the phonograph" on two measures (OCR lines 4196–4205). *Others off-text.*

Q32 RI.3.2 TRAP

D

Main Idea: a good invention both inspires new inventions and changes how we live, and the light bulb does both best (OCR lines 3701–3711, 4254–4256). **Trap:** A/B/C are stray details.

Worksheet C — IB Critical Thinking · Answer Key

5 questions · PYP Key Concepts · DOK 2–4

Q1 Perseverance DOK 3

In **"Timeless Thomas,"** show a PROBLEM or failure Edison faced, what he DID about it, and the RESULT for the world.

MODEL ANSWER

A Problem or Failure: Thomas Edison's inventions did not always work the first time. He had thousands of failures. He once said, "I know several thousand things that won't work." For example, he built his nickel-iron storage battery to power an electric car, but the electric car did not catch on. **What Edison Did:** Edison did not quit. He treated his failures as a "valuable" and necessary part of inventing, and he always tried again. When the electric car failed, he found many other uses for the battery. He gathered a team of scientists and started the world's first research lab, the Invention Factory, and kept experimenting. **The Result:** Edison's patience gave the world the phonograph, the motion picture camera, the storage battery, and the incandescent light bulb — inventions we still use today. He was awarded 1,093 patents. **How perseverance helped:** Because Edison never gave up after a failure, he was able to keep improving his ideas until they worked. His "valuable failures" show that not quitting is part of making a successful invention.

KEYWORDS (3)

problem = many failures (electric car) · action = try again, valuable failures · result = phonograph, light bulb, 1,093 patents

SCORING NOTES

- Full credit (3): a problem/failure + what Edison did + result + how never giving up made a difference.
- Partial (2): problem and action named but no result.
- Minimal (1): only "he was an inventor" with no problem or result.

TEACHING STEPS

Step 1 · Three boxes in order.

강사: *"Label three boxes: Problem, Action, Result. What goes in each?"*

코멘트. 상자 세 칸 — 문제·행동·결과. 각 칸에 들어갈 내용 적기.

Step 2 · One failure.

강사: *"Underline one failure Edison had and what he did next."*

코멘트. 에디슨의 실패 하나와 그다음 한 일에 밑줄.

Step 3 · Perseverance sentence.

강사: *"Finish: 'Edison kept inventing by ____.'"*

코멘트. 문장 완성 — '에디슨은 ____함으로써 계속 발명했다.'

STUDENT PROMPTS

- "Why did Edison call his failures 'valuable'?"

에디슨은 왜 자신의 실패를 '소중하다'고 했을까?

- "What might have happened if he had quit after one failure?"
  한 번 실패하고 그만뒀다면 어떻게 됐을까?
- "When did YOU keep trying after something went wrong?"
  무언가 잘 안됐을 때 너는 언제 계속 시도했니?

DIFFERENTIATION

STRUGGLING Put 3 cards (something failed / he tried again / he made a great invention) in order.
카드 3장 — 실패·재시도·발명 성공 순서 배열.

ADVANCED Compare Edison's view of failure to Aunt Rose's words in Rosie Revere.
에디슨의 실패관을 Rosie Revere의 로즈 할머니 말과 비교.

COMMON MISCONCEPTIONS

PITFALL "Edison's inventions all worked the first time." No — he had thousands of failures first.
에디슨 발명이 처음부터 다 된 게 아님 — 수천 번 실패 후 성공.

PITFALL "Edison worked completely alone." He led a large team at the Invention Factory.
에디슨이 혼자 다 한 게 아님 — Invention Factory에서 팀을 이끎.

LEARNER PROFILE LINK

RISK-TAKERS Trying again after thousands of failures takes courage.
수천 번 실패 후에도 다시 시도 — Risk-taker.

THINKERS Using a failure to learn what won't work is real thinking.
실패에서 안 되는 것을 배움 — Thinker.

KOREAN BRIDGE

표현 "An inventor who never gives up can change the world." → "포기하지 않는 발명가는 세상을 바꿀 수 있다."
끈기(Perseverance) 핵심 문장.

단어 "failure / 실패 — invention / 발명 — breakthrough / 돌파구 — patent / 특허"
핵심 단어 영-한.

Q2 Change DOK 3

In **"A Bumpy Ride,"** trace THREE steps in how the bicycle changed, then explain how each inventor built on the one before.

MODEL ANSWER

Step 1 · Early Bike: The first machine was Karl von Drais's draisienne. It had no pedals and was more like a sit-on scooter that riders pushed with their feet. It was a start, but it was hard to ride on rough roads. **Step 2 · An Improvement:** Later, Kirkpatrick Macmillan added foot pedals that turned the back wheel, so riders no longer had to push the ground. Then Pierre Michaux moved the pedals to the front wheel. The big-wheeled "ordinaries" went faster, and finally the Rover "safety" bike added a chain and gears so the wheel did not need to be huge. **Step 3 · Modern Bike:** John Boyd Dunlop's air-filled tire made the ride smooth, and later riders added good helmets for safety. The modern bicycle was fast, safe, and comfortable. **How each inventor built on the last:** Each inventor looked at the problems of the bike before and fixed one of them — adding pedals, then gears, then air-filled tires. No single person made the modern bike; many small inventions together produced one successful design.

KEYWORDS (3)

early = draisienne, no pedals | improve = pedals, gears, "safety" bike | modern = air tires, helmets, safe & fast

SCORING NOTES

- Full credit (3): all three steps (early → improvement → modern) + how each inventor built on the last.
- Partial (2): three steps named but no "built on the one before" idea.
- Minimal (1): only "bikes got better" with no steps.

TEACHING STEPS

Step 1 · Three boxes on a timeline.

강사: *"Label three boxes: Early, Improvement, Modern. What goes in each?"*

코멘트. 상자 세 칸 — 초기·개선·현대. 각 칸 채우기.

Step 2 · One fix.

강사: *"Write one problem an inventor fixed (pedals, gears, or tires)."*

코멘트. 발명가가 고친 문제 하나 적기(페달·기어·타이어).

Step 3 · Built-on sentence.

강사: *"Finish: 'Each inventor made the bike better by ____.'"*

코멘트. 문장 완성 — '각 발명가는 ____함으로써 자전거를 개선했다.'

STUDENT PROMPTS

- "Why was the draisienne hard to ride?"
  드라이지엔은 왜 타기 어려웠을까?
- "Why did the 'ordinaries' have such a big front wheel?"
  '오디너리'는 왜 앞바퀴가 그렇게 컸을까?
- "How is improving a bike like improving any invention?"

자전거를 개선하는 일이 다른 발명을 개선하는 것과 어떻게 닮았을까?

DIFFERENTIATION

STRUGGLING Put 3 picture cards (scooter bike / big-wheel bike / modern bike) in time order.

그림 카드 3장 — 스쿠터·큰바퀴·현대 자전거 순서 배열.

ADVANCED Explain why the author calls the history "a bumpy ride."

저자가 역사를 왜 '울퉁불퉁한 여정'이라 부르는지 설명.

COMMON MISCONCEPTIONS

PITFALL "One person invented the whole bicycle." No — many inventors each added one improvement.

한 사람이 자전거를 다 만든 게 아님 — 여러 발명가가 하나씩 개선.

PITFALL "The first bikes had pedals." The draisienne had no pedals at all.

첫 자전거에 페달이 있던 게 아님 — 드라이지엔은 페달이 없었음.

LEARNER PROFILE LINK

INQUIRERS Each inventor asked, "How can I make this better?"

각 발명가가 '어떻게 더 낫게 만들까' 질문 — Inquirer.

KNOWLEDGEABLE Knowing the older bike helped the next inventor improve it.

이전 자전거를 알았기에 다음 발명가가 개선 — Knowledgeable.

KOREAN BRIDGE

표현 "Many small inventions, one after another, can make one great invention." → "작은 발명이 차례로 모여 하나의 훌륭한 발명을 만든다."

변화(Change) 핵심 문장.

단어 "contraption / 기계 장치 — pedal / 페달 — improve / 개선하다 — design / 설계"

핵심 단어 영-한.

Q3 Perseverance DOK 3

In **"Rosie Revere, Engineer,"** show how Rosie felt at first, what her aunt taught her, and how Rosie changed.

MODEL ANSWER

How Rosie Felt After the Crash: When Rosie's heli-o-cheese-copter flew for a moment and then crashed, she thought she had failed. She felt "baffled, embarrassed, perplexed." She believed she would never try to build anything again and would never be a great engineer. She thought the laughing meant her aunt was making fun of her. **What Aunt Rose Taught Her:** But Great-Great-Aunt Rose hugged Rosie and said, "You did it! Hooray! It's the perfect first try!" She explained that the copter "did just what it needed to do" — before it crashed, it flew. She told Rosie that the only true failure is to quit, and she handed Rosie a notebook to keep inventing. **How Rosie Changed:** Rosie smiled "as it all became clear." She kept inventing, and she taught the kids in her class to cheer for their "perfect failures." **Why this matters for an inventor:** Rosie learned that a flop is part of inventing, not the end. An inventor must keep trying after a failure, just as Edison did, because the only real failure is giving up.

KEYWORDS (3)

at first = baffled, wanted to quit

aunt's lesson = "perfect first try," don't quit

change = keeps inventing, cheers failures

SCORING NOTES

- Full credit (3): how Rosie felt + what Aunt Rose taught + how Rosie changed + why it matters for an inventor.
- Partial (2): feeling and lesson named but no change.
- Minimal (1): only "her copter crashed" with no lesson.

TEACHING STEPS

Step 1 · Before and after.

강사: *"Make two boxes: How Rosie felt at first / How Rosie felt after her aunt spoke."*

코멘트. 두 칸 — 처음 로지의 마음 / 할머니 말 이후의 마음.

Step 2 · The wise words.

강사: *"Underline the line where Aunt Rose says the flop was a success."*

코멘트. 로즈 할머니가 실패를 성공이라 말하는 문장에 밑줄.

Step 3 · Lesson sentence.

강사: *"Finish: 'Rosie learned that the only true failure is ____.'"*

코멘트. 문장 완성 — '로지는 진짜 실패란 ____뿐임을 배웠다.'

STUDENT PROMPTS

- "Why did Rosie think she had failed?"
  로지는 왜 자신이 실패했다고 생각했을까?
- "How can a crash still be a 'perfect first try'?"
  추락한 것이 어떻게 '완벽한 첫 시도'일 수 있을까?
- "What would you say to a friend whose invention flopped?"

발명이 실패한 친구에게 너는 뭐라고 말해 주겠니?

DIFFERENTIATION

STRUGGLING Match 2 cards ("sad, wants to quit" / "happy, keeps trying") to before and after.

카드 2장 — '슬픔·포기'와 '기쁨·재시도'를 전·후에 매칭.

ADVANCED Explain how Rosie's lesson is the same as Edison's in Timeless Thomas.

로지의 교훈이 Timeless Thomas의 에디슨과 어떻게 같은지 설명.

COMMON MISCONCEPTIONS

PITFALL "The copter never flew, so it was a total failure." It DID fly for a moment before it crashed.

헬리콥터가 전혀 못 난 게 아님 — 추락 전 잠깐 날았음.

PITFALL "Aunt Rose was laughing to be mean." She was happy and proud, not mean.

로즈 할머니가 비웃은 게 아님 — 기쁘고 자랑스러워서 웃음.

LEARNER PROFILE LINK

RISK-TAKERS Building and testing an invention that might flop takes courage.

실패할 수 있는 발명을 만들고 시험 — Risk-taker.

REFLECTIVE Rosie thinks again about what "failure" really means.

로지가 '실패'의 뜻을 다시 생각 — Reflective.

KOREAN BRIDGE

표현 "The only true failure is to quit." → "진짜 실패는 포기하는 것뿐이다."

끈기(Perseverance) 핵심 문장.

단어 "engineer / 공학자 — flop / 실패작 — perplexed / 당황한 — try again / 다시 시도하다"

핵심 단어 영-한.

Q4 Innovation DOK 3

Compare HOW **Thomas Edison** in Timeless Thomas and **Rosie Revere** in Rosie Revere, Engineer each kept inventing.

MODEL ANSWER

Thomas Edison · A Real Inventor: Timeless Thomas is a true biography. Edison was a real inventor who had thousands of failures but treated them as a "valuable" part of inventing. He never stopped trying, and his persistence gave the world the phonograph, the light bulb, and many other inventions. **Rosie Revere · A Story Inventor:** Rosie Revere, Engineer is a made-up narrative poem. Rosie is a story character, a shy girl who builds gadgets. When her cheese-copter crashed, she wanted to quit — until her aunt showed her that the flop was really a "perfect first try" and that the only true failure is quitting. **What both inventors show:** Even though one is a real person and one is a poem, both stories teach the same lesson about inventing: failure is part of the work, and a successful inventor keeps trying. Edison said he knew "several thousand things that won't work," and Rosie learned to cheer for her "perfect failures." **So:** A biography and a poem can carry the same big idea. Both show that it takes a brilliant idea AND the courage to keep going after failure to make a successful invention.

KEYWORDS (3)

Edison = real, "valuable failures" | Rosie = story, "perfect first try" | both: keep trying after failure

SCORING NOTES

- Full credit (3): how EACH kept inventing (real life / poem) + the same lesson both show about failure.
- Partial (2): both described but only one "keep trying" idea.
- Minimal (1): only "both invented things" with no comparison.

TEACHING STEPS

Step 1 · T-chart.

강사: *"Make a T-chart: Edison / Rosie. How did each keep inventing?"*

코멘트. T-차트 — 에디슨 / 로지. 각자 어떻게 계속 발명했나?

Step 2 · Real or story?

강사: *"Mark which one is a true biography and which is a made-up poem."*

코멘트. 실제 전기인 쪽과 지어낸 시인 쪽 표시.

Step 3 · Same lesson sentence.

강사: *"Finish: 'Both Edison and Rosie show that ____.'"*

코멘트. 문장 완성 — '에디슨과 로지 모두 ____임을 보여 준다.'

STUDENT PROMPTS

- "How are Edison's words about failure like Aunt Rose's?"
  에디슨의 실패에 대한 말이 로즈 할머니 말과 어떻게 닮았니?
- "Can a made-up poem teach a true lesson?"
  지어낸 시도 참된 교훈을 줄 수 있을까?
- "Which inventor do you feel closer to, and why?"

두 발명가 중 누가 더 가깝게 느껴지니? 왜?

DIFFERENTIATION

STRUGGLING Match 2 cards ("real person" / "story character") to the right selection.

카드 2장을 알맞은 선택지와 매칭.

ADVANCED Add A Bumpy Ride as a third example of "keep improving."

A Bumpy Ride를 '계속 개선하기'의 셋째 예로 추가.

COMMON MISCONCEPTIONS

PITFALL "Only one of them is about inventing." Both are about inventors and failure.

둘 중 하나만 발명 이야기인 게 아님 — 둘 다 발명가와 실패에 관함.

PITFALL "A poem can't be true." A poem is fiction, but it can still teach a real lesson.

시가 사실일 수 없다는 건 오해 — 허구라도 참된 교훈을 줌.

LEARNER PROFILE LINK

THINKERS Comparing a biography and a poem for one idea is real thinking.

전기와 시를 하나의 생각으로 비교 — Thinker.

OPEN-MINDED Seeing the same lesson in two very different texts is open-minded.

아주 다른 두 글에서 같은 교훈을 봄 — Open-minded.

KOREAN BRIDGE

표현 "A real life and a story poem can teach the same lesson about inventing." → "실제 삶과 이야기 시가 발명에 대해 같은 교훈을 줄 수 있다."

혁신(Innovation) 핵심 문장.

단어 "biography / 전기 — narrative poem / 이야기 시 — failure / 실패 — persistence / 끈기"

핵심 단어 영-한.

Q5 Innovation / Perseverance DOK 4

Essential Question: What does it take to make a successful invention? Use ideas from TWO of the selections in this module to answer.

MODEL ANSWER

It takes a brilliant idea, hard work, and the courage to keep trying after failure to make a successful invention. In **"Timeless Thomas,"** Thomas Edison had thousands of failures, but he treated them as a "valuable" part of inventing and always tried again. His persistence gave the world the phonograph, the light bulb, and many other inventions we still use. In **"Rosie Revere, Engineer,"** Rosie built a flying machine that crashed, and she wanted to quit. But her aunt showed her that the flop was a "perfect first try," because it flew before it crashed, and that the only true failure is quitting. **My Big Idea:** A successful invention takes more than one bright idea — it takes a person who does not give up when something goes wrong. Edison kept inventing after thousands of failures, and Rosie kept inventing after her copter crashed. As Thomas Edison said, "To invent, you need a good imagination and a pile of junk" — and the patience to keep trying.

KEYWORDS (3)

Edison = "valuable failures," never quits

Rosie = "perfect first try," keeps going

Big Idea: idea + hard work + don't quit

SCORING NOTES

- Full credit (4): two selections named + what each shows about inventing + WHY it matters + a Big Idea answering the EQ.
- Partial (3): two selections + ideas + Big Idea weak.
- Developing (2): one selection only.
- Minimal (1): generic ("inventing is hard") without naming a selection.

SYNTHESIS FRAME

SELECTION 1 IDEA	CONNECTING WORD	SELECTION 2 IDEA
Name what the selection shows about inventing: Timeless Thomas = never give up after failure; A Bumpy Ride = build on others' ideas; Rosie Revere = a flop is a first try; Edison's Best Invention = a good invention inspires more and changes life.	BOTH / AND — joins two ideas about inventing (a brilliant idea AND never quitting, or one inventor building on another) into one Big Idea answering the EQ.	An idea from a DIFFERENT selection than Selection 1 — so the two show two parts of what it takes to invent.

REAL-WORLD CONNECTION

KOREA 우리 주변의 발명도 한 번에 되지 않는다 — 시제품을 여러 번 고치는 발명가, 실수를 고쳐 더 나은 앱을 만드는 개발자처럼. 에디슨의 끈기, 로지의 '완벽한 첫 시도'처럼 실패를 딛고 계속 시도하는 것이 성공의 열쇠다.

'실패를 딛고 다시 시도한' 사례를 떠올리기.

PERSONAL What is something you tried, failed, and want to try again?

네가 시도했다 실패했지만 다시 해 보고 싶은 일은 무엇이니?

RUBRIC

LEVEL	WHAT IT LOOKS LIKE
Exceeding (4)	Names 2 selections + what each shows about inventing (idea, hard work, trying again) + WHY it matters + a Big Idea answering the EQ.
Meeting (3)	Names 2 selections + what each shows + a Big Idea.
Approaching (2)	Names 2 selections but only retells the plot — no idea about inventing named.
Beginning (1)	Generic ("inventing is fun") without text evidence.

Watch out: Many G3 students will just retell what happened (Edison made a light bulb; Rosie's copter crashed) and stop. Push them to name what it TAKES to invent — a brilliant idea, hard work, building on others, and the courage to keep trying after failure — and to say WHY it matters. The full EQ asks WHAT IT TAKES to make a successful invention, not just what happens in the story.

TEACHING STEPS

Step 1 · Two selection cards.

강사: *"Pick 2 selections. For each, write what it shows about inventing."*

코멘트. 선택지 둘 — 각자 발명에 대해 무엇을 보여 주는지 적기.

Step 2 · Why it matters.

강사: *"Under each, write WHY that idea matters for making an invention."*

코멘트. 각각 아래 — 그 생각이 발명에 왜 중요한지.

Step 3 · Big Idea sentence.

강사: *"Write one sentence answering the EQ. Use both selections."*

코멘트. EQ에 답하는 한 문장 — 두 선택지 모두 사용.

STUDENT PROMPTS

- "Is one good idea enough, or do you also need to keep trying?"
  좋은 아이디어 하나면 충분할까, 아니면 계속 시도도 필요할까?
- "Which inventor's story stuck with YOU the most?"
  어느 발명가의 이야기가 너에게 가장 남았니?
- "Why is failing an important part of inventing?"
  실패가 왜 발명의 중요한 부분일까?

DIFFERENTIATION

STRUGGLING Sentence frame: "In ____, ____ shows that inventing takes ____. In ____, ____ shows that inventing takes ____. Both show that a successful invention needs ____."

3단 문장 골격.

ADVANCED Use THREE selections and sort them by what they show (a brilliant idea / building on others / never quitting).

선택지 셋 사용 — 보여 주는 것별 분류(아이디어·계승·끈기).

COMMON MISCONCEPTIONS

PITFALL "A great invention just needs one smart idea." It also needs hard work and trying again after failure.

훌륭한 발명에 똑똑한 아이디어 하나면 된다는 건 오해 — 노력과 재시도도 필요.

PITFALL "Retelling the plot answers the EQ." The EQ asks WHAT IT TAKES to invent, not just what happens.

줄거리 요약은 EQ 답이 아님 — '무엇이 필요한지'를 물음.

LEARNER PROFILE LINK

THINKERS Synthesizing two selections into one Big Idea about inventing is the heart of critical thinking.

두 선택지를 한 Big Idea로 종합 — Thinker.

RISK-TAKERS Explaining why inventors must try again shows you value taking risks.

발명가가 다시 시도해야 하는 이유를 설명 — Risk-taker.

KOREAN BRIDGE

표현 "A successful invention takes a brilliant idea and the courage to keep trying after failure." → "성공적인 발명에는 빛나는 아이디어와 실패 후에도 계속 시도하는 용기가 필요하다."

M8 마무리 문장 — 학생 글쓰기 핵심.

인용 "To invent, you need a good imagination and a pile of junk." — Thomas Edison → "발명을 하려면 좋은 상상력과 잡동사니 더미가 필요하다."

에피그래프 — 상상과 끈기.

Worksheet D — MAP Practice · Answer Key

20 items · RIT 195 → 215 · Lexile 410L–680L

MAP Mini-Passage Word Counts (Worksheet D)

- Passage 1 · The Sticky Note · **152 words** · Lexile 410–460L · RIT 195–201
- Passage 2 · The Better Pencil · **159 words** · Lexile 470–530L · RIT 202–207
- Passage 3 · How the Bicycle Bell Rang · **160 words** · Lexile 540–600L · RIT 208–212
- Passage 4 · The Boy Who Made Braille · **161 words** · Lexile 610–680L · RIT 213–215

All four passages within target band 120–200 words; RIT 195→215; Lexile 410L–680L. Word counts match the workbook passage meta exactly. Each passage models a different invention — the sticky note, the eraser pencil, the bicycle bell, and Braille. Passages are NEW — no recall of the four anchor selections.

PASSAGE 1 · THE STICKY NOTE

Q1 · RIT 195

A

Lexile 410L · Key Detail. "A scientist named Spencer was trying to make a very strong glue." *Distractors B/C/D name a book, a light bulb, a toy.*

Q2 · RIT 197

B

Lexile 420L · Key Detail. "the little papers that marked the pages in his book kept falling out." *Trap A/C/D off-text.*

Q3 · RIT 199

C

Lexile 440L · Cause & Effect. "the paper could stick to a page and peel off without a tear." *Trap A/B/D off-text.*

Q4 · RIT 200

D

Lexile 450L · Vocabulary. A "failure" is something that does not work the way you wanted. *Trap A/B/C give the wrong picture.*

Q5 · RIT 201

B

Lexile 460L · Main Idea. "even a mistake can become a useful invention" when you see a problem in a new way. *Trap A/C/D narrow or off-topic.*

PASSAGE 2 · THE BETTER PENCIL

Q6 · RIT 202

B

Lexile 470L · Key Detail. "Her eraser was always rolling off the desk or hiding in her bag." *Trap A/C/D off-text.*

Q7 · RIT 204

D

Lexile 490L · Key Detail. "she fixed the eraser to the top of her pencil with the ring." *Trap A/B/C off-text.*

Q8 · RIT 205

A

Lexile 500L · Cause & Effect. "the eraser was always right there... She never had to search for it again." *Trap B/C/D off-text.*

Q9 · RIT 206

C

Lexile 510L · Inference. "Her friends saw her clever pencil and wanted one too" — the invention was useful to others. *Trap A/B/D miss it.*

Q10 · RIT 207

B

Lexile 530L · Main Idea. "She had taken two things that already existed and joined them to solve a real problem." *Trap A/C/D narrow it.*

PASSAGE 3 · HOW THE BICYCLE BELL RANG

Q11 RIT 208

C

Lexile 540L · Key Detail. "People walking did not hear the bicycles coming. There were many small crashes." *Trap A/B/D off-text.*

Q12 RIT 209

A

Lexile 560L · Cause & Effect. "a shout was easy to miss in the noise of the city." *Trap B/C/D off-text.*

Q13 RIT 210

D

Lexile 570L · Key Detail. "a tiny hammer struck the dome and made a bright ring" when the rider pressed a lever. *Trap A/B/C off-text.*

Q14 RIT 211

B

Lexile 580L · Inference. The rider could "press it with a thumb without letting go of the bike" — so they could warn people and still steer safely. *Trap A/C/D miss it.*

Q15 RIT 212

D

Lexile 600L · Main Idea. "a small idea, but it solved a real problem and helped many people stay safe." *Trap A/B/C trivialize it.*

PASSAGE 4 · THE BOY WHO MADE BRAILLE

Q16 RIT 213

A

Lexile 610L · Key Detail. "blind students could read only a few books... heavy and hard to make." *Trap B/C/D off-text.*

Q17 RIT 214

C

Lexile 630L · Key Detail. "a secret code of raised dots that soldiers used to send messages in the dark." *Trap A/B/D off-text.*

Q18 RIT 214

B

Lexile 650L · Vocabulary. A "brilliant" idea is one that is very clever and smart. *Trap A/C/D give the wrong picture.*

Q19 RIT 215

D

Lexile 670L · Cause & Effect. "A blind reader could feel the dots with a fingertip and read quickly." *Trap A/B/C off-text.*

Q20 RIT 215

C

Lexile 680L · Main Idea. "One determined young person changed millions of lives with a simple, clever idea." *Trap A/B/D trivialize it.*